

7 + 8

Calculus I Paper 1

FIRST PRINCIPLES AND BASIC RULES

QUESTION 1

1.1 Determine the derivative of $f(x) = 1 - 2x^2$ by using first principles. (4)

1.2 Determine the following derivatives:

1.2.1 $\frac{dy}{dx}$ if $y = \frac{1}{2}x^4 + bx$ (2)

1.2.2 $D_x \left[\frac{5x^3 - 6\sqrt{x^3} + 5}{x} \right]$ (4)

1.3 Given $f(x) = -\frac{x^2}{2} + x$. A tangent to the graph of f has a gradient of -5 and x intercept $(a; 0)$. Determine the value of a . (5)

QUESTION 2

2.1 Given that $f(x) = 2x^3$. Determine $f'(x)$ from first principles. (5)

2.2 Determine the derivatives of the following:

2.2.1 $y = \frac{2}{3x^5}$ (1)

2.2.2 $y = \frac{x-25}{\sqrt{x}-5}$ (3)

QUESTION 3

3.1 If $f(x) = \frac{4}{x}$, determine $f'(x)$ from first principles. (5)

3.2 Determine:

3.2.1 $\frac{dy}{dx}$ if $y = 5x^2 + 5x + 2$ (2)

3.2.2 $D_x \left[\sqrt[3]{x^2} - \frac{1}{2}x \right]$ (3)

3.3 Given: $p(x) = x^3 + 2x$

Show, using relevant calculations, why it is not possible for a tangent drawn to the graph of p to have a negative gradient. (3)

CUBIC FUNCTIONS

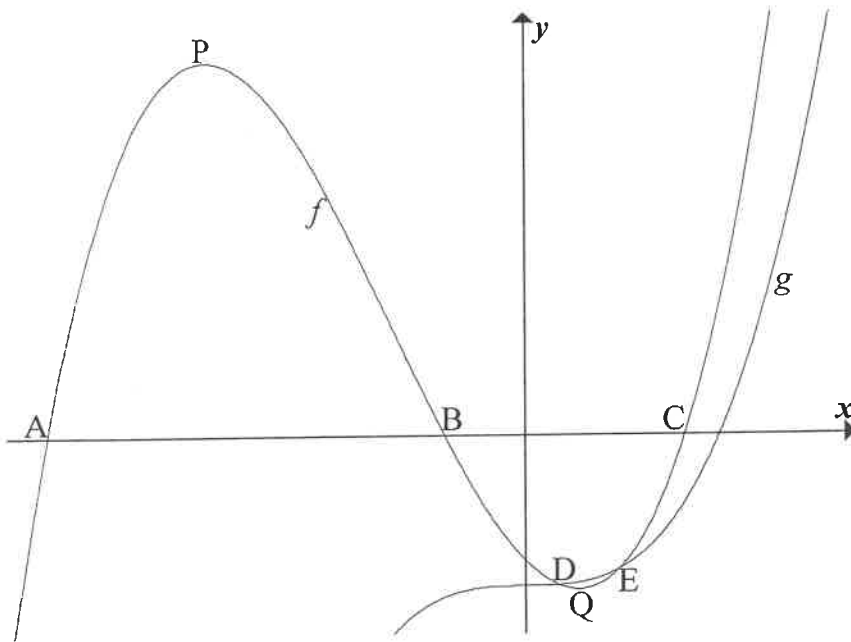
QUESTION 4

Given: $f(x) = -x^3 + x^2 + 8x - 12$

- 4.1. Calculate the x -intercepts of the graph of f . (5)
- 4.2 Calculate the coordinates of the turning points of the graph of f . (5)
- 4.3 Sketch the graph of f , showing clearly all the intercepts with the axes and turning points. (3)
- 4.4 Write down the x -coordinate of the point of inflection of f . (2)
- 4.5 Write down the coordinates of the turning points of $h(x) = f(x) - 3$. (2)

QUESTION 5

The sketch shows the graphs of $f(x) = x^3 + 5x^2 - 8x - 12$ and $g(x) = x^3 - 15$. P and Q are the stationary points of f . D and E are the points of intersection of f and g .



- 5.1 Determine the co-ordinates of A, B and C. (4)
- 5.2 Determine the co-ordinates of P and Q. (6)
- 5.3 Determine the x - co-ordinates of D and E. (5)
- 5.4 Determine the x - co-ordinate of the point of inflection of f . (3)
- 5.5 The graph of g has two tangents parallel to the line $y - 24x = 5$. Determine the equation of the one that touches the graph of g at a point with a positive x - co-ordinate. (5)
- 5.6 For which values of p will the equation $f(x) = p$ have exactly two distinct roots? (2)

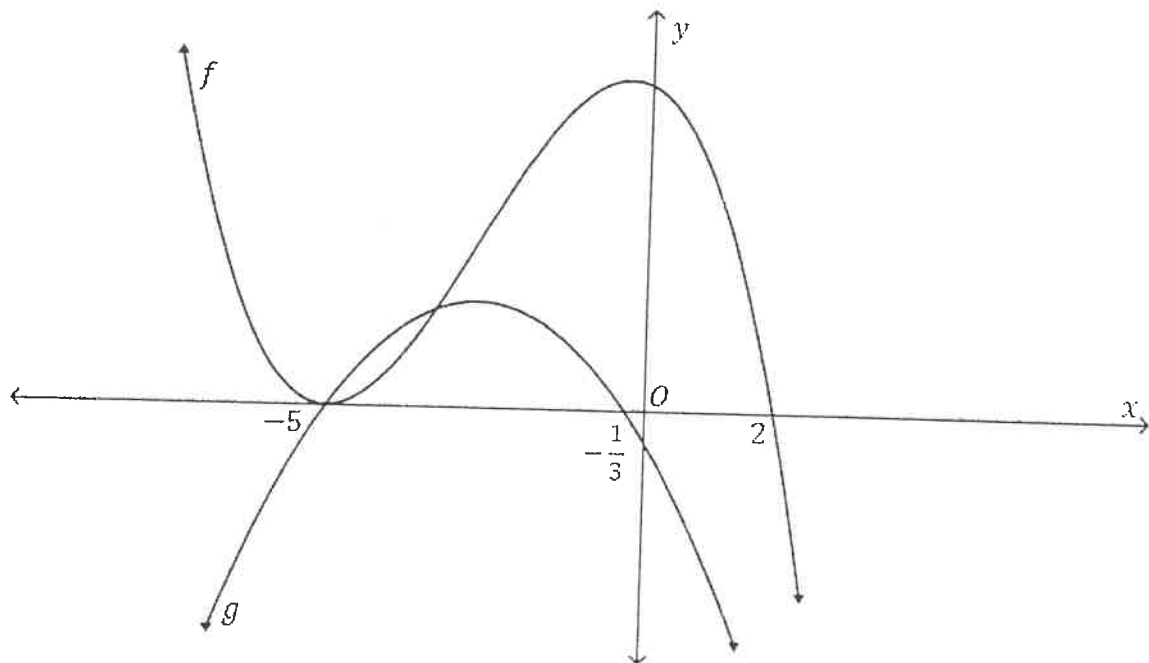
QUESTION 6

The sketch shows the graphs of the following functions:

$$f(x) = ax^3 + bx^2 - 5x + 50$$

$$g(x) = f'(x) = px^2 + qx + t$$

The x -intercepts of f are $(-5; 0)$ and $(2; 0)$. The x -intercepts of g are $(-5; 0)$ and $(-\frac{1}{3}; 0)$.

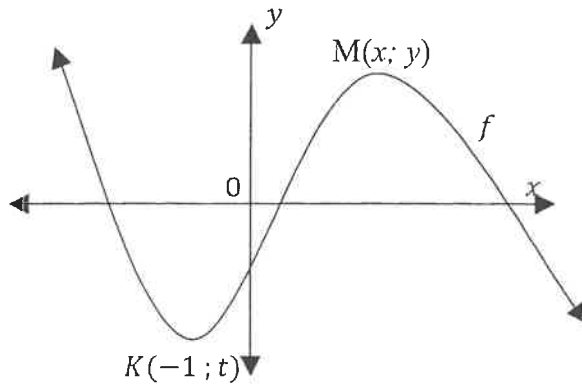


Use the given information and sketch to answer the following questions:

- 6.1 Determine the value of $g[f(2)]$. (2)
- 6.2 Calculate the average gradient of f over the interval $x \in [-5; 0]$. (2)
- 6.3 Determine the values of x for which $f''(x) < 0$. (2)

QUESTION 7

The function f below has equation $f(x) = -x^3 + 3x^2 + cx - 4$, with $K(-1; t)$ and $M(x; y)$ as the turning points.



- 7.1 Show that $c = 9$. (4)
- 7.2 Determine the coordinates of $M(x; y)$. (4)
- 7.3 Show that $f'(x) - 12 \leq 0$ for all real values of x . (5)

QUESTION 8

Let $f(x) = ax^3 + bx^2 + cx$.

- 8.1 Show that the x -co-ordinates of the stationary points of f are given by
$$x = \frac{-b \pm \sqrt{b^2 - 3ac}}{3a}. \quad (3)$$
- 8.2 Derive a formula, in terms of a , b and/or c for the x -co-ordinate of the point of inflection of f . (3)
- 8.3 State how many stationary points f will have in each of the following cases. Simply write: **TWO**, **ONE** or **NONE**.
- 8.3.1 $a = 1$, $b = 3$ and $c = 3$ (1)
- 8.3.2 $a = b = c$ and $b \neq 0$ (1)

[8]

Calculus 1: Solutions

First Principles and Basic Rules

Question 1:

$$\begin{aligned}
 1.1. \quad f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1 - 2(x+h)^2 - (1 - 2x^2)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1 - 2(x^2 + 2xh + h^2) - 1 + 2x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1 - 2x^2 - 4xh - 2h^2 - 1 + 2x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-4xh - 2h^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{h(-4x - 2h)}{h} \\
 &= \lim_{h \rightarrow 0} (-4x - 2h) \\
 &= -4x - 2(0) \\
 &= \underline{-4x}
 \end{aligned}$$

$$\begin{aligned}
 1.2.1. \quad y &= \frac{1}{2}x^4 + bx \\
 \frac{dy}{dx} &= \underline{2x^3 + b}
 \end{aligned}$$

$$\begin{aligned}
 1.2.2. \quad D_x \left[\frac{5x^3 - 6\sqrt{x^3} + 5}{x} \right] \\
 = D_x \left[\frac{5x^3}{x} - \frac{6\sqrt{x^3}}{x} + \frac{5}{x} \right]
 \end{aligned}$$

(2)

$$\begin{aligned}
 &= D_x \left[5x^2 - 6x^{\frac{3}{2}} + 5x^{-1} \right] \\
 &= D_x \left[5x^2 - 6x^{\frac{1}{2}} + 5x^{-1} \right] \\
 &= \underline{10x - 3x^{-\frac{1}{2}} - 5x^{-2}}
 \end{aligned}$$

1.3.

$$\begin{aligned}
 f'(x) &= -x + 1 = -5 \\
 x &= 6
 \end{aligned}$$

$$\begin{aligned}
 f(6) &= -\frac{(6)^2}{2} + 6 \\
 &= \underline{-12}
 \end{aligned}$$

$$\begin{aligned}
 y - y_1 &= m(x - x_1) \\
 y + 12 &= -5(x - 6) \\
 y + 12 &= -5x + 30 \\
 y &= \underline{-5x + 18}
 \end{aligned}$$

$$\begin{aligned}
 -5x + 18 &= 0 \\
 x &= \underline{18/5}
 \end{aligned}$$

$$\therefore a = 18/5$$

Question 2

2.1.

$$\begin{aligned}
 f(x+h) &= 2(x+h)^3 \\
 &= 2(x+h)(x^2 + 2xh + h^2) \\
 &= 2(x^3 + 3x^2h + 3xh^2 + h^3) \\
 &= 2x^3 + 6x^2h + 6xh^2 + 2h^3
 \end{aligned}$$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2x^3 + 6x^2h + 6xh^2 + 2h^3 - 2x^3}{h} \\
 &= \lim_{h \rightarrow 0} \frac{6x^2 + 6xh + 2h^2}{1} \\
 &= 6x^2 + 6x(0) + 2(0)^2 \\
 &= \underline{6x^2}
 \end{aligned}$$

2.2.1.

$$\begin{aligned}
 y &= \frac{2}{3x^5} \\
 &= \frac{2}{3} x^{-5}
 \end{aligned}$$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{2}{3} (-5x^{-6}) \\
 &= \underline{\underline{-\frac{10}{3} x^{-6}}}
 \end{aligned}$$

2.2.2.

$$\begin{aligned}
 y &= \frac{x-25}{\sqrt{x}-5} \\
 &= \frac{(\sqrt{x}-5)(\sqrt{x}+5)}{(\sqrt{x}-5)} \\
 &= \sqrt{x} + 5 \\
 &= x^{\frac{1}{2}} + 5
 \end{aligned}$$

$$\frac{dy}{dx} = \underline{\underline{\frac{1}{2} x^{-\frac{1}{2}}}}$$

Question 3.1

$$\begin{aligned}
 3.1. \quad f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \left(\frac{4}{x+h} - \frac{4}{x} \right) \div h \\
 &= \lim_{h \rightarrow 0} \frac{4x - 4(x+h)}{x(x+h)} \times \frac{1}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4x - 4x - 4h}{x(x+h)} \times \frac{1}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-4h}{x(x+h)} \times \frac{1}{h} \\
 &= \frac{-4}{x(x+0)} \\
 &= \frac{-4}{x^2}
 \end{aligned}$$

3.2.1.

$$\begin{aligned}
 y &= 5x^2 + 5x + 2 \\
 \frac{dy}{dx} &= 10x + 5
 \end{aligned}$$

3.2.2.

$$\begin{aligned}
 D_x \left[\sqrt[3]{x^2} - \frac{1}{2}x \right] \\
 &= D_x \left[x^{2/3} - \frac{1}{2}x \right] \\
 &= \frac{2}{3}x^{-1/3} - \frac{1}{2}
 \end{aligned}$$

3.3.

$$\begin{aligned}
 p(x) &= x^3 + 2x \\
 p'(x) &= 3x^2 + 2 > 0 \quad \text{For all } x \in \mathbb{R}.
 \end{aligned}$$

Question 4:

4.1.

$$-x^3 + x^2 + 8x - 12 = 0$$

$$x^3 - x^2 - 8x + 12 = 0$$

$$(x-2)(x^2 + x - 6) = 0$$

$$(x-2)(x+3)(x-2) = 0$$

$$\underline{x=2}, \quad \text{or} \quad \underline{x=-3}$$

4.2.

$$f'(x) = -3x^2 + 2x + 8 = 0$$

$$3x^2 - 2x - 8 = 0$$

$$(3x+4)(x-2) = 0$$

$$x = -\frac{4}{3}$$

$$\text{or} \quad \underline{x=2}$$

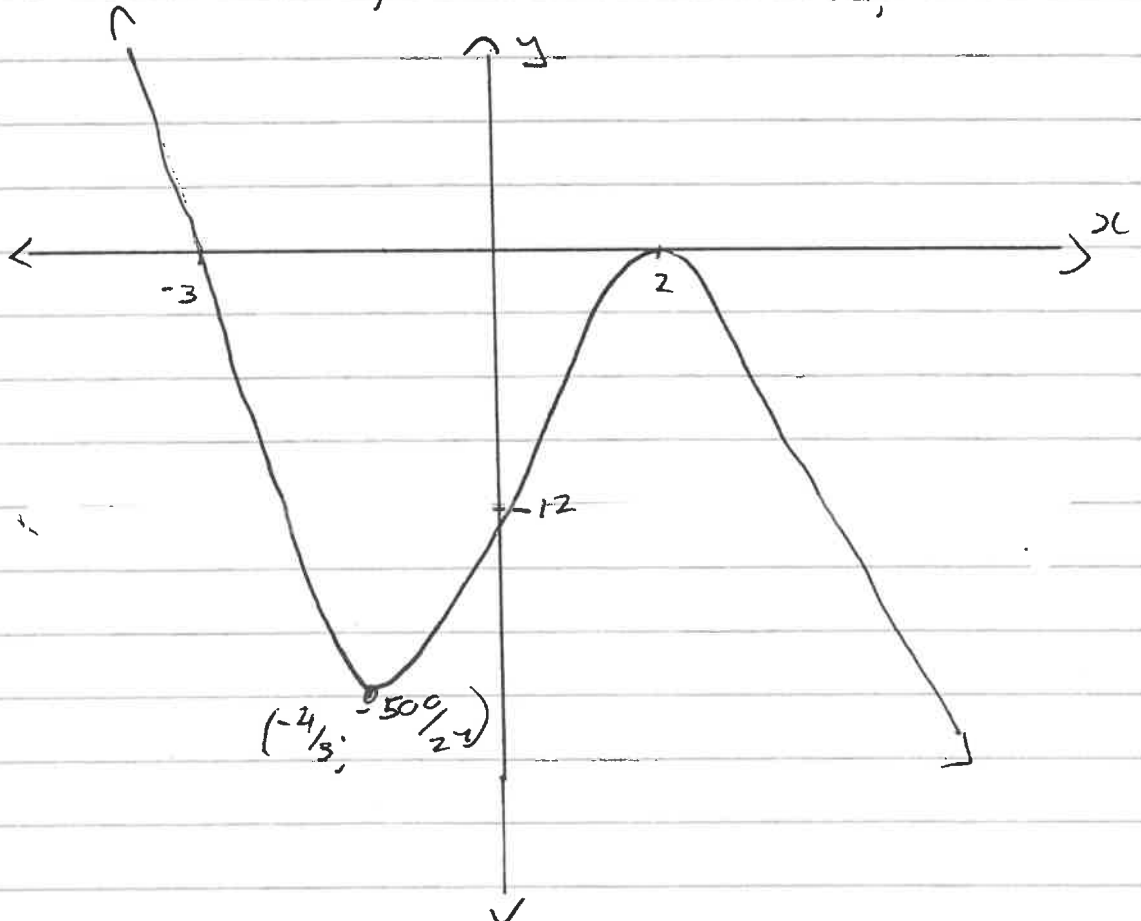
$$\underline{\underline{-\frac{4}{3}}}$$

$$f\left(-\frac{4}{3}\right) = -\left(-\frac{4}{3}\right)^3 + \left(-\frac{4}{3}\right)^2 + 8\left(-\frac{4}{3}\right) - 12$$

$$= \underline{\underline{-\frac{500}{27}}}$$

Coordinates $\left(-\frac{4}{3}, -\frac{500}{27}\right)$ and $(2, 0)$

4.3



(6)

4.4. $f''(x) = -6x + 2 = 0$
 $x = \frac{1}{3}$

4.5. $(-\frac{4}{3}, -\frac{58}{27})$ and $(2, -3)$

Question 5:

5.1. $x^3 + 5x^2 - 8x - 12 = 0$
 $(x-2)(x^2 + 7x + 6) = 0$
 $(x-2)(x+6)(x+1) = 0$
 $x = 2$, or $x = -6$, or $x = -1$

A $(-6, 0)$

B $(-1, 0)$

C $(2, 0)$

5.2. $f'(x) = 3x^2 + 10x - 8 = 0$
 $(3x-2)(x+4) = 0$
 $x = \frac{2}{3}$, or $x = -4$

$f(\frac{2}{3}) = (\frac{2}{3})^3 + 5(\frac{2}{3})^2 - 8(\frac{2}{3}) - 12$
 $= -\frac{400}{27}$

$f(-4) = (-4)^3 + 5(-4)^2 - 8(-4) - 12$
 $= 36$

$P(-4, 36)$ and $Q(\frac{2}{3}, -\frac{400}{27})$

5.3

$$x^3 + 5x^2 - 8x - 12 = x^3 - 15$$

$$5x^2 - 8x + 3 = 0$$

$$(5x - 3)(x - 1) = 0$$

$$\underline{x = \frac{3}{5}}, \quad \text{OR} \quad \underline{x = 1},$$

5.4

$$f''(x) = 6x + 10 = 0$$

$$x = \frac{10}{6}$$

$$\underline{x = \frac{5}{3}},$$

5.5

$$y - 24x = 5$$

$$y = 24x + 5$$

$$\therefore \underline{m = 24},$$

$$g'(x) = 3x^2 = 24$$

$$x^2 = 8$$

$$x = \pm 2\sqrt{2}$$

$$\therefore x = 2\sqrt{2} \quad \text{sub into } g$$

$$g(2\sqrt{2}) = (2\sqrt{2})^3 - 15$$

$$= \underline{16\sqrt{2} - 15},$$

$$y - y_1 = m(x - x_1)$$

$$y - 16\sqrt{2} + 15 = 24(x - 2\sqrt{2})$$

$$y - 16\sqrt{2} + 15 = 24x - 48\sqrt{2}$$

$$\underline{y = 24x - 32\sqrt{2} - 15},$$

5.6

$$\underline{p = 36}$$

$$\text{OR } \underline{p = \frac{-400}{27}},$$

Question 6:

$$\begin{aligned}
 6.1. \quad & g[f(2)] \\
 &= g(0) \\
 &= \underline{-5}
 \end{aligned}$$

$$\begin{aligned}
 6.2. \quad m &= \frac{0 - 50}{-5 - 0} \\
 &= \underline{10}
 \end{aligned}$$

$$6.3. \quad \underline{x > -8/3}$$

Question 7:

$$\begin{aligned}
 7.1. \quad f'(x) &= -3x^2 + 6x + c \\
 f'(-1) &= 0 \\
 -3(-1)^2 + 6(-1) + c &= 0 \\
 -3 - 6 + c &= 0 \\
 \underline{c = 9}
 \end{aligned}$$

$$\begin{aligned}
 7.2. \quad f'(x) &= -3x^2 + 6x + 9 = 0 \\
 x^2 - 2x - 3 &= 0 \\
 (x - 3)(x + 1) &= 0 \\
 \underline{x = 3} \quad \text{or} \quad \underline{x = -1}
 \end{aligned}$$

$$\begin{aligned}
 f(3) &= -(3)^3 + 3(3)^2 + 9(3) - 4 \\
 &= \underline{23}
 \end{aligned}$$

$$\underline{M(3, 23)}$$

7.3

$$\begin{aligned}
 & -3x^2 + 6x + 9 - 12 \\
 & = -3x^2 + 6x - 3 \\
 & = -3(x^2 - 2x + 1) \\
 & = -3(x-1)^2
 \end{aligned}$$

$$\begin{aligned}
 (x-1)^2 & \geq 0 & \text{For all } x \in \mathbb{R} \\
 \therefore -3(x-1)^2 & \leq 0 & \text{For all } x \in \mathbb{R}
 \end{aligned}$$

Question 8:

$$8.1. \quad f'(x) = 3ax^2 + 2bx + c = 0$$

$$\begin{aligned}
 x &= \frac{-(2b) \pm \sqrt{(2b)^2 - 4(3a)(c)}}{2(3a)} \\
 &= \frac{-2b \pm \sqrt{4b^2 - 12ac}}{6a} \\
 &= \frac{-2b \pm 2\sqrt{b^2 - 3ac}}{6a} \\
 &= \frac{-b \pm \sqrt{b^2 - 3ac}}{3a}
 \end{aligned}$$

$$8.2. \quad f''(x) = 6ax + 2b = 0$$

$$6ax = -2b$$

$$x = \frac{-2b}{6a}$$

$$x = -\frac{b}{3a}$$

8.3.1. One

8.3.2. none